Activity 1, Intermediate - Code & Go Mouse

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In this adventure, students will learn coding skills with Colby, the programmable robot mouse from APH. The students will program the sequence of steps and Colby will race to find the cheese.

Setup - Activity 1

This activity can take several sessions. Students will need time to create the maze before solving it.

Objectives

- Students will be able to learn basic coding skills.
- Students will be able to understand the basic components of computational thinking.

Materials Needed

- APH's Code & Go Robot Mouse
- Assorted textured paper
- Stickers
- Printable 5" x 5" squares
- Scissors
- Glue Stick
- Other art supplies

Ideas for Carrying Out This Activity

- Work in some route planning and O&M skills to your adventure.
- If students are having trouble identifying right and left according to the mouse's perspective. Try recreating the grid with foam mat floor tiles and have the student pretend to be Colby. The student can walk through the life-size course to get a better understanding of the mouse's perspective.

- Allowing students to arrange these tiles to create their own grid, shape or "road" encourages the student to be creative. Taking the tactile diagram and "building" the diagram is another form of creativity. An advanced student could take this a step farther by creating their own Code and Go course then create their version of the accompanying Activity Card using graph paper and tactile materials.

Adventure Map: Activity 1

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: How does developing a route for Colby(the mouse) relate to writing code?
  - Review vocabulary with your student:
  - Algorithmic Thinking - a way of approaching problems that involves breaking them down into smaller, more manageable parts. It is a process that involves identifying the steps needed to solve a problem and then implementing those steps in a logical and efficient manner.
  - Debug - identify and remove errors from a computer program.

2. Introduce Code & Go

- Let the student explore all the components of the Code & Go set.
- Let them figure out how the tiles fit together.
- Let them try to control Colby the mouse before telling them how it works.
- Let them guess the purpose of the other parts.

3. Practice Together

- Demonstrate how the robot mouse, Colby, can be programmed by using the buttons.
- Set up a simple maze, show the student where Colby should start and where they should end up (at the cheese).
- Have the student plan out Colby's route using the arrow cards.
- Have the student program Colby and see if their program works.
- If something goes wrong, have the student try to debug their program, go back through their initial plan with the arrow cards and see if they can figure out the step where things went wrong.
- After the maze has been solved, have the student try another, harder maze.

- You can use the maze walls to create traps and dead-ends and bridges to guide Colby in a certain direction.

4. **Design Colby's Mouse Town**

   - To give your student additional opportunities to explore the idea of creating a town for Colby and be creative. Use the 5" x 5" square (this is the size of the tiles from Code and Go) and have your student draw out/ design the places in the town. Some examples are a School, Library, Park and Cheese Shop. Make sure to reiterate sketching techniques covered from the introductory adventures. They can use markers, stickers, textured paper, braille labels or any other art supplies you have.

5. **Sharing and Discussion**

   - What did we learn today?
   - How is giving Colby directions like writing a computer program?
   - How important do you think debugging is in programming?

6. **Conclusion**

   - Summarize the key points of the adventure.
   - Have the student reflect on what went well and what was challenging.
   - If time allows, have the students create their own mazes.

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Activity 2, Intermediate - Colby's Mouse Town

Activity 2, Intermediate - Colby's Mouse Town

Download this activity

In this adventure, students will learn introductory programming concepts by planning and navigating routes through Colby's town on graph paper. They will practice sequencing by breaking bigger tasks into smaller steps.

Setup - Activity 2

This activity can be broken down into several elements for lesson planning.

Objectives

- Understand that programs execute commands in sequence
- Break problems down into sequential steps
- Represent sequences and loops tactually

Materials Needed

- Graph paper with 1" grid
- Assorted Texture Paper
- Stickers
- Braille labels
- Printable block coding commands
- Scissors
- Glue stick
- Other art supplies
- Lego blocks
- Wheatley Tactile Diagramming Kit

Ideas for Carrying Out This Activity

- Encourage students to work with partners if possible

Adventure Map: Activity 2

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: How can we break bigger tasks down into smaller steps? How can breaking down a larger problem into smaller problems make it easier to come up with a solution?
  - Review vocabulary with your student:
  - Algorithm - A list of steps that you can follow to finish a task
  - Sequence - In a program each step is an instruction to be performed. Sequence is the order in which the steps are carried out.
  - Loop - A sequence of instructions that is repeated a certain number of times.
- Show examples of simple programs drawn on graph paper.

2. Introduce Colby's Mouse Town

- For the introduction, reflect on the last adventure where students programmed Colby to navigate different paths to get to the cheese.
- For this exercise, we will use sheets of graph paper with a 1" grid to plan a layout for Colby's town. Starting at the upper left-hand corner, make this spot Colby's house. This will give you a starting spot for the next section. Have the student come up with other locations in the town, use places from the last activity, some places might be:
  - School
  - Library
  - Park
  - Bank
  - Cheese Shop
- Have the student indicate where each place is located on the grid.
  - They can use markers, stickers, textured paper, braille labels or any other art supplies you have.

3. Navigating the Town

- Go over the possible command you can give to Colby, see the printable block coding commands.
- Give your student a problem: Colby needs to go to school, what is the quickest route he can take?
  - Start out with an easy challenge and give your student harder and harder challenges.

  - Have some roadblocks or other obstacles you can add to the grid to change Colby's path.
  - After a couple routes, introduce the repeat command.

4. **Sharing and Discussion**

   - What did we learn today?
   - What if we have a bunch of the same commands in a row, what can we do to simplify our list of commands?

5. **Conclusion**

   - Summarize the key points of the lesson.
   - If time allows, have the students create another town or expand on the one they have. They could also create a grid for the inside of Colby's house or the inside of the school.
   - You can also ask that students incorporate one or more loops in their algorithm.

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Activity 3, Intermediate - Drawing with Colby

Activity 3, Intermediate - Drawing with Colby

Download this activity

In this adventure, students will learn introductory programming concepts by planning and using Colby's path to create shapes. Students will be introduced to perspective and using coordinates on a coordinate grid.

Setup - Activity 3

This activity can be broken down into several elements for lesson planning.

Objectives

- Understand that a coordinate grid system is used to describe the position of object on that grid
- Represent an ordered pair as a point on a coordinate grid

Materials Needed

- Coordinate grid with quadrants (.prn file)
- Coordinate grid with quadrants (.pdf file)
- Graph paper with 1" grid, (.prn file)
- Graph paper with 1" grid, (.pdf file)
- Stickers
- Wiki Sticks
- Markers
- Ruler
- Wheatley Tactile Diagramming Kit

Ideas for Carrying Out This Activity

- Encourage students to work with partners if possible
- Use legos to help with the concept of coordinate grids. Use the lego baseplate as the grid and each of the round studs can represent a point where two lines intersect.

Adventure Map: Activity 3

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: Why is the skill of plotting and interpreting points on a coordinate grid valuable in creating computer graphics?
  - Review vocabulary with your student:
  - Coordinate Grid - A coordinate plane is a two-dimensional plane formed by the intersection of two perpendicular lines. Think of the coordinate plane as a large flat surface, like a tabletop or the floor. It's a space where we can locate different points.
  - X-axis - The horizontal line that divides the top and bottom of the coordinate grid.
  - Y-axis - The vertical line that divides the left and right of the coordinate grid.
  - Origin - The point at which the x and y axis intersect.
  - Ordered Pair - An x and y coordinate given within parentheses and separated by a comma. The x value always comes first. The coordinates of a point (x, y) describe how far you've moved horizontally (x) and vertically (y) from your starting point. If you move two steps to the right (x = 2) and three steps forward (y = 3), your position would be (2, 3).
  - Quadrants - The coordinate plane is divided into four equal parts by the intersection of the x-axis (the horizontal line) and the y-axis (the vertical line). These four regions are called quadrants because they each represent one-quarter of the whole coordinate plane. quadrant 1 is the upper right, quadrant 2 is the upper left, quadrant 3 is the lower left and quadrant 4 is the lower right.
- In computer graphics, the screen is represented only by quadrant 4. The origin, or (0,0) point is in the upper left corner. Since there is no other quadrant used in computer graphics the y values are positive. If we were in math class and using the coordinate grid system, a point in quadrant 4, all y values would be negative.
- Show an example of a full coordinate grid and just the grid of quadrant 4 that will be used for this activity.

2. Perspective

- Visual perspective refers to how objects appear relative to each other based on their position and distance from the viewer.

- To understand perspective, imagine looking straight down at Colby's path on the coordinate grid. From this bird's eye view directly overhead, the sizes and positions of Colby and the grid all look proportional and accurate.
- Now imagine zooming out, so you are viewing Colby's path from farther away. As you zoom out, Colby will appear smaller and smaller on the grid. The distances between Colby's path points will seem closer together.
- If we zoomed out so far that Colby was tiny, we could represent his position with just a dot, rather than his full image. This demonstrates how visual perspective changes how we perceive size, distance, and position as our viewpoint changes.

3. Plotting Points

- Plotting points on a coordinate grid involves identifying the location of a point based on its coordinates (x, y) on the grid. Here are the steps to plot points on a coordinate grid:
  - Identify the Coordinates: Each point on the grid is represented by a pair of coordinates (x, y). The x-coordinate tells you how far to the right the point is from the origin. The y-coordinate tells you how down the point is from the origin.
  - Plot the Point: Start at the origin (0, 0), which is in the upper left corner. Move horizontally (right) the number of units specified by the x-coordinate. Move vertically (down) the number of units specified by the y-coordinate. Place a mark or sticker at the intersection of the horizontal and vertical lines to represent the point.
- Example: If you have a point with coordinates (3, 4), you would start at the origin (0,0) and move three units to the right along the x-axis and then four units down and mark the point at that intersection.
- Additional Tips: The order of the coordinates matters. X is always first, Y is always second (x, y).
- The Y values are not negative, even though we are in quadrant 4. This is different when using coordinate grids in math class.
- We are counting the intersections of lines and marking points where two lines intersect. We're not using the open space between the lines.

4. Drawing with Colby's Path

- We are going to start out with a rectangle. Colby is going to start on point (2,2). Start at the origin (0,0), in the upper left corner and move 2 lines to the right. Now move 2 lines down and mark that point. Next we're going to move to the 2nd point, (5,2). Mark the new point and mark Colby's path, you can use a marker or wiki stix. We will repeat this process for points 3 and 4. Point 3 is (5,3), mark the point and Colby's path to that point. Point 4 is (2,3), mark the point and Colby's path to that point. And the last move is to return to the starting point, (2,2).

- Colby has just drawn his first shape!
- We will call this rectangle: (2,2), (5,2), (5,3), (2,3) and (2,2)
- What shape will tracing the path between these points create?
  - (1,3), (5,3), (5, 7), (1, 7) and (1,3)
- What shape will tracing the path between these points create?
  - (8, 1), (8, 4), (5, 4) and (8, 1)
- Now you try to draw a simple shape and list the points your shape uses.

5. **Sharing and Discussion**

- Have students share the shapes they created by tracing Colby's path between points.
- Ask students to explain the ordered pairs they used and how they plotted each point.
- Discuss how changing the coordinates affects the shape that is drawn. For example:
  - What happens if you increase or decrease the x or y values?
- How can you make the shape bigger or smaller?
- Have students share any challenges they faced in visualizing or plotting the points. Provide guidance as needed.
- Discuss real world applications of using coordinate grids, such as:
  - Mapping locations using latitude and longitude
  - Designing objects in CAD (computer assisted design) programs
  - Designing digital graphics

6. **Conclusion**

- A coordinate grid is used to precisely specify locations of points using x and y values.
- Ordered pairs denote the x and y coordinates that identify a unique point on the grid.
- By plotting points and tracing the path between them, we can create different shapes.
- Perspective affects how we perceive size and distance of objects.
- Coordinate grids are useful for mapping positions, designing objects, and creating graphics.
- Review vocabulary such as coordinate plane, x-axis, y-axis, origin, ordered pair, quadrant.
- Explain how today's adventure serves as an introduction to working with coordinates, which is important for programming graphics using SVG code.

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Activity 4, Intermediate - Code Quest

Activity 4, Intermediate - Code Quest

In this adventure, students will learn coding skills with CodeQuest, the free iPad app from APH. The students will program solutions to help the astronaut return to their spaceship.

Setup - Activity 4

Download this activity

This activity can take several sessions. There are 30 levels within the CodeQuest app. Not all levels need to be completed. The levels start out easy and get harder as the student progresses. Students have the opportunity to work with tactile maps, 3D objects, and technology, ensuring a multi-sensory learning experience that caters to different learning styles and abilities. They are encouraged to take their time, be creative, and experiment with different solutions, fostering a sense of ownership over their code.

Objectives

- Students will be able to learn basic coding skills.
- Students will be able to understand that debugging or problem solving is part of the programming process.

Materials Needed

- iPad or iPhone
- APH CodeQuest free app (Links to App Store)
- (For braille readers) Tactical Maps of levels are very helpful while using the app
- (Optional) 3D print files for Alien, Arrows, Astronaut and Rocket Ship
- 3D printer files and grids for embossing.

Ideas for Carrying Out This Activity

- Focus on iPad skills throughout this adventure
  - Tapping

    - Swiping
    - Scanning
    - Selecting
    - VoiceOver

Adventure Map: Activity 4

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Using embossed Grid 1, challenge students to create "code" to move the astronaut to the rocket ship using the following commands. They must use each command at least once!
    - Left, Up, Down, Right
    - Consider using a small, 3D object for the students to move around the grid for trial and error.
- Any combination that results with the arrival of the astronaut to the rocket ship is acceptable.
- Next, have the student write down the commands (in order) that would be used to tell the astronaut how to follow a path to get to the rocket ship.
- If time, students can check their work by reading code out loud and having someone else follow commands to be sure that the code does not have any "bugs." Edit as necessary.

2. Kratos (1st Planet, 6 levels)

- Introduce the student to the symbols used for the maps, either on the screen or using the tactile maps. If using the tactile maps, be sure the student takes time to look at both the key and the map.
- Have the student share how they think the astronaut will have to move to get to the rocket ship. If the student has difficulty keeping track of the steps, it may help to have the student write down the steps first.
- Introduce the student to the buttons at the bottom of the screen.
- Have the student select the correct steps and then select the Play button to have the astronaut perform the steps.
- Encourage the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
- Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.

- After the student has completed all 6 levels, they can select Planet Helios to go to the next Planet.

3. **Helios (2nd Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The alien is new in this planet.
   - Let the student know that they only have to land on the square with the alien to pick it up, but they must go to each square with an alien before arriving at the rocket ship.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Attis to go to the next Planet.

4. **Attis (3rd Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The breakable wall is new in this level.
   - Let the student know that they cannot move to the square with the breakable wall until they go to a square next to the breakable wall and then use the blaster to remove it.
   - Have the student find the new Blaster button at the bottom.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Castor to go to the next Planet.

5. **Castor (4th Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The stronger breakable wall is new in this level.
   - Let the student know that they will need to look for the number on the stronger breakable wall to find out how many times they will need to use the blaster to remove it.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Pollux to go to the final Planet.

6. **Pollux (5th Planet, 6 levels)**
   - Have the student find the new Loop button at the bottom.
   - Show the student how they can select the Loop button and then swipe up or down to change the number of times to loop the command.
   - Placing this Loop command after another command makes that command repeat for the designated number of times.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship. The Loop will help decrease the number of steps.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.

7. **Conclusion**
   - Summarize the key points of the adventure.
   - Have the student reflect on what went well and what was challenging.

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